

UIJONG YANG

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SUMMARY

- Backend and cloud software engineer who deep-dives into root causes through measurement and improves on that evidence.
- Optimized core API latency by up to **42%** at Boeing Korea and built **750+ automated tests (76% coverage)** from scratch at PT Korea.
- Active contributor to large open-source projects, including Uber Cadence (Go) and Youki (Rust container runtime).

SKILLS

Languages	Python, Go, Rust, C#, SQL
Backend	FastAPI, ASP.NET
Cloud Native	AWS, Terraform, Docker
CI	GitHub Actions, GitLab CI
AI Tooling	Claude Code, Codex
Certifications	AWS Advanced Networking (ANS), AWS Solutions Architect Associate (SAA), TOEIC Speaking IH

EXPERIENCE

Boeing Korea, Cloud System Software Engineer Intern

Oct 2025 - Mar 2026

C#, Python, ASP.NET, EF, SQL Server, Docker | Seoul, Korea

- **Core API Optimization:** Isolated unavoidable US-Korea network latency via ping tests, profiled EF-generated SQL with SET STATISTICS, converted subqueries to JOIN/EXISTS, added composite indexes, and cut DB round-trips. **Table-view query 42% (2046ms → 1189ms), single-record detail query 41% (1389ms → 823ms).**
- **Graph UI Rendering:** Slimmed server-side DTOs and moved Dagre data-structure transforms to the server so the browser only renders. **Eliminated 5+ minute graph display delays (10+ on slower internal networks)** caused by payload and client-side transform overhead. **Reduced graph rendering compute time by ~20% (JavaScript-based measurement).**
- **AI Model CPU Fallback:** Traced the call stack of a GPU (CUDA)-only model and reimplemented an equivalent CPU-based Python fallback. **Removed GPU dependency** and deployed into CPU-only production and development environments.

PT Korea, Python Developer Intern

Jun 2025 - Sep 2025

FastAPI, Pytest, PostgreSQL, OpenCV, PaddleOCR | Seoul, Korea

- **Test Automation:** In a codebase with no tests, traced every function's call stack into a dependency diagram and built unit/integration tests bottom-up from the lowest-dependency modules. **750+ tests, 76% coverage from zero**, blocking regressions and improving release stability.

OPEN SOURCE CONTRIBUTIONS

Cadence, Distributed Workflow Orchestration Engine (Uber)

Go

- Replaced a custom rate limiter with the Go standard library (rate.Limiter), reducing complexity (#7585); reproduced a negative-RPS permanent deadlock via a self-written POC and added validation.
- Improved task matching: removed redundant sync retries (#7650), propagated context timeouts (#7709), added select-based timeouts with async re-matching (#7827) and dynamic Adaptive TaskList Scaler config (#8035). **5 PRs merged.**

Youki, Container Runtime

Rust

- Fixed bugs and added unit tests (#3602, #3608, #3631); analyzed tty inode validation (fstat/fstatfs) as a CVE-2025-52565 (TOCTOU) defense and published a technical write-up before contributing.

EDUCATION

Soongsil University, B.S. Computer Science

Mar 2022 - Feb 2027

GPA 3.79 / 4.5 | All graduation requirements completed. Available for full-time internship

PROJECTS

Single Sign-On Service (ITSUPPORT-AUTH)

Ongoing

Go, AWS Cognito, Lambda, API Gateway, SQS | Soongsil University, ITSsupport Committee

- Built an OIDC SSO on AWS serverless; an API Handler proxies token exchange/refresh, logout, and onboarding so clients never call Cognito directly.
- **Async event pipeline:** Decoupled login synchronization as Post Authentication → SQS → Consumer, with idempotency via SQL transactions and unique constraints. **Cut login latency 72% (637ms → 176ms).**
- **Cold-start mitigation:** Rewrote the auth system from Python to Go and applied Provisioned Concurrency. **~124ms average API Handler latency**, sustained with no reported service interruptions during a two-week scholarship application period (Grafana Cloud).
- **Load test (K6):** Verified a **120 RPS** throughput ceiling (Cognito auth quota, zero Lambda/API GW throttles) with excess load rejected in ~8ms. A forced 5-min Consumer/DB outage left login p50 unchanged (<0.5%). **22K backlogged messages auto-recovered with zero loss.**

AWS Infrastructure Management (ITSUPPORT-AWS)

Ongoing

Terraform, AWS, CloudFormation, GitHub Actions | Soongsil University, ITSsupport Committee

- Designed and provisioned a 3-tier AWS infrastructure as code with Terraform and CloudFormation, gated by checkov/tflint pre-commit hooks.
- **Cost optimization:** Consolidated dev instances onto a single ARM64 (Graviton) EC2 Spot instance and removed idle EC2/RDS resources. **44.4% lower monthly cost (\$203.32 → \$112.93).**

Container Runtime (from scratch)

Personal Project

Rust, Linux namespaces, cgroups v2, pivot_root

- Implemented process/mount/network isolation, CPU/memory limits, and root-filesystem isolation **from scratch** with Linux namespaces, cgroups v2, and pivot_root.

RAG Agent System

Personal Project

Python, FastAPI, ChromaDB, gpt-4o-mini

- Ran **33 controlled experiments** (hypothesize → implement → measure → analyze logs) on a Korean-law multiple-choice (KMMLU) RAG system; discovered zero-shot prompts outperform context-injected ones (60-64% vs 44-47%) and raised accuracy **51.58% → 53.28% (dev set).**